Synthesis Report: LFU Cache Controller `rtl-v2-final` on NanGate45

NanGate45, not TSMC; pre-layout; area = sum of cell areas; no power analysis

Adarsh
Department of Electrical and Electronics Engineering
BITS Pilani, K K Birla Goa Campus, Goa, India

Pravin Mane
Department of Electrical and Electronics Engineering
BITS Pilani, K K Birla Goa Campus, Goa, India

1. Scope and limits: read this first

What this report is. Logic synthesis (Yosys and ABC) and static timing analysis (OpenSTA) of the LFU cache controller of tag `rtl-v2-final`, mapped to the NanGate45 open cell library, with functional verification of the resulting netlists. Every number is taken from committed result files through `results/P7/TABLES.md` (commit `f0f82ce`) and checked by `tools/paper_numbers_check.py`.

What it is not.

- **Not TSMC.** NanGate45 is an open library for a generic predictive process. This report supersedes an earlier Word report whose TSMC 28nm area, power and timing figures have no source file; none of them is used here.
- **Pre-layout.** Interconnect is estimated by the liberty wire-load model; the clock is ideal (no clock tree); there is no placement, routing or parasitic extraction.
- **Area** is the sum of the liberty cell areas, not a placed area. Storage is built from flip-flops; there is no SRAM macro.
- **No power analysis** was run, so this report contains no power numbers.
- **Gate-level simulation is zero-delay** (no SDF), with cell models generated by Yosys from the liberty file; formal equivalence is proven at depth 16 only.
- **Flow noise.** Synthesizing identical logic in different textual forms moves Fmax by 1.96–16.62% (earlier flow) and 4.92–16.23% (current flow); differences smaller than that are not evidence of a design effect.

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2 Design

RTL. `rtl/v2/lfu_cache_controller.v` at tag `rtl-v2-final` (commit `7705a7b`). The file in the repository is identical to the tag (checked by `tools/make_tables.py`).

Parameters. Address width 20 bits, vector width 128 bits, counter width 6 bits (saturating at 63), 16 groups for victim selection. Synthesized at depths (`CACHE_DEPTH`) 16, 64 and 256 lines.

Counter modes (parameter `COUNTER_MODE`). Mode 0, the default, sets the count of a newly installed tag to one; mode 1 lets the new tag continue from the count left in the slot. Mode 2 (per-address history) is not implemented: elaboration stops with an error. Both implemented modes are synthesized.

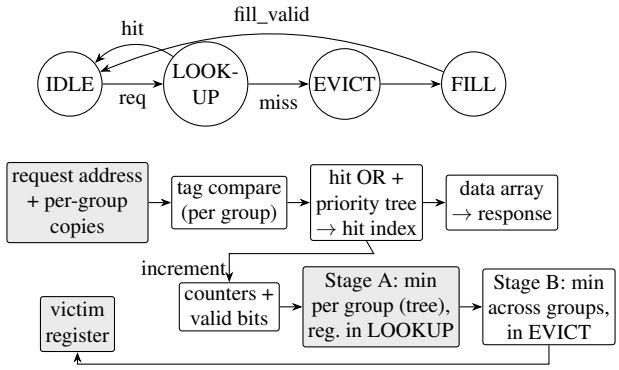


Figure 1: The LFU controller: FSM (top) and datapath (bottom; shaded blocks are registers). A hit returns to IDLE; a miss selects the victim (Stage A registered in LOOKUP, Stage B in EVICT) and installs the fill in FILL.

Operation (Fig. 1). A request moves the FSM from IDLE to LOOKUP. The tag compare runs per group on registered copies of the request address; a per-group OR and a balanced priority tree give the lowest matching line (the *hit path*). On a hit the line's counter is incremented and the vector is returned. On a miss the victim is the line with the smallest count: **Stage A** is a reduction tree inside each group, registered at the end of LOOKUP; **Stage B** is a reduction tree across the 16 group minima, evaluated in EVICT, whose result is written to the victim register; an invalid line counts as smaller than any valid one, and ties go to the lower slot index. FILL writes the returned vector, the tag and the new count into the victim slot.

3 Toolchain and reproducibility

Item	Version / identifier
Yosys	0.33, git `2584903a060`
ABC	the copy built into Yosys 0.33 (no separate version recorded)
OpenSTA	3.1.0, commit `7b261909c1730d963252142e136daaa0199e333c`
CUDD (OpenSTA dependency)	tag `3.0.0`, commit `f54f533303640afd5dbe47a05ebeabb3066f2a25`
Liberty	`NangateOpenCellLibrary_typical.lib`, corner typical, 25C, 1.1V
source	OpenROAD-flow-scripts, commit `6c6fc7e21b2d0b5b88b1a52d0a5860ea4d88e9e5`
sha256	`8d540a4d4cf6d09d27c87ad067857a9c0c2eeb023ab7a56e058cd3113db4e9b1`
size	6,692,032 bytes (not redistributed; see `results/P5/SOURCES.md` for its licence header)
Icarus Verilog	12.0 (local), 11.0 (cluster)

Sources: `results/P5/SOURCES.md` (build record of the local tools), the cluster gate-level logs and `results/P5f_gls/jobs.txt`; all collected in `results/P7/TABLES.md`, section "Flow settings and toolchain". Flow scripts: `synthesis/p5/` (synthesis template, ABC constraints, STA script, driver), `synthesis/p5e/` (timing-driven ABC script and template, multicycle STA), `synthesis/p5c/`, `synthesis/p5d/` (design-iteration drivers), `synthesis/p5f/` (gate-level simulation). The commands that regenerate every table are listed in the appendix.

4 Flow

Synthesis (`synthesis/p5/synth.ys.in`): read the liberty as a cell library, read the RTL, set depth and counter mode, `proc`, copy the `keep` attribute onto the flip-flops that drive the per-group address copies (otherwise Yosys merges the identical registers), `synth -flatten`, `dfflibmap` to the liberty flip-flops, `abc` technology mapping, tie cells, `stat -liberty` (cell counts and area) and `check -assert`.

ABC. The earlier flow (P5–P5d) calls `abc -liberty -D <period> -constr`, i.e. Yosys's built-in script. The current flow (P5e, `synthesis/p5e/abc_td.script.in`) runs three delay-oriented restructuring passes (`&st; &syn2; &if -g -K 6; &synch2; &nf -D`), then buffering including flip-flop outputs (`buffer -c -p`) and gate sizing (`upsize -c`, `dnsize -c`), all with the liberty's wire-load model. Its delay target is the clock period minus 141.5 ps: ABC sees flip-flop outputs as inputs and flip-flop inputs as outputs, so clock-to-Q and setup are subtracted beforehand; 141.5 ps is clk→Q 0.0983 ns plus setup 0.0432 ns on the worst path of the earlier flow's depth-16, mode-0, 1 GHz run.

Constraints. Inputs driven by a BUF_X1 cell and outputs loaded with 3.795620 fF, the same in ABC and in STA.

STA (`synthesis/p5/sta.tcl`). Wire-load model 5K_hvratio_1_1; ideal clock; input and output delay 0 relative to the clock; the asynchronous reset's recovery and removal paths are excluded (`set_false_path -from rst_n`), because the reset net has no buffer tree before layout; its slew and capacitance violation is still reported but not counted below. Multicycle exceptions are used only in the cross-check of Section 9; all headline numbers use none.

Fmax $= 1/(T - \mathrm{WNS})$ from the worst setup path of the netlist synthesized for a period T; "met" means WNS ≥ 0. Each configuration is synthesized twice, for 1 GHz and for 500 MHz.

5 Results

One table per design iteration. Cells, flip-flops (FFs), area, flip-flop share of area and Fmax are from the 1 GHz netlist; "WNS 500 MHz" is from the netlist synthesized for 500 MHz; "Viol." counts nets or pins that violate a max-slew or max-capacitance limit in the 1 GHz / 500 MHz netlists (reset net excluded). Values are rounded for display (area and Fmax to integers, slack to 3 decimals); the unrounded values are in `results/P7/TABLES.md`.

Table 1: P5 baseline (tag `rtl-v2-baseline`, commit `0d66a3a`): serial minimum and priority loops; earlier flow. The 1 GHz and 500 MHz netlists are identical at every depth (the ABC target did not bind).

Depth	Mode	Cells	FFs	Area (μm^2)	FF share of area	Fmax (MHz)	WNS 1 GHz (ns)	WNS 500 MHz (ns)	Viol. 1G / 500M
16	0	9,399	2,657	23,689	0.597	203	−3.923	−2.923	0 / 0
16	1	10,467	2,657	24,643	0.574	215	−3.649	−2.649	2 / 2
64	0	36,605	10,131	90,869	0.593	195	−4.136	−3.136	0 / 0
64	1	37,204	10,131	91,623	0.588	191	−4.249	−3.249	2 / 2
256	0	143,139	39,925	358,573	0.592	167	−4.975	−3.975	18 / 18
256	1	144,521	39,925	364,405	0.583	165	−5.048	−4.048	130 / 130

Table 2: P5b (commit `4debe84`): Stage-B reduction tree and per-group address copies; earlier flow.

Depth	Mode	Cells	FFs	Area (μm^2)	FF share of area	Fmax (MHz)	WNS 1 GHz (ns)	WNS 500 MHz (ns)	Viol. 1G / 500M
16	0	10,381	2,957	25,879	0.608	629	−0.589	met	0 / 0
16	1	10,690	2,957	26,137	0.602	624	−0.602	met	1 / 1
64	0	40,818	10,431	96,163	0.577	476	−1.100	−0.272	17 / 19
64	1	41,303	10,431	97,302	0.570	473	−1.116	−0.295	2 / 2
256	0	143,583	40,225	359,160	0.596	188	−4.307	−3.307	64 / 64
256	1	147,232	40,225	362,078	0.591	185	−4.400	−3.400	83 / 83

Table 3: P5c (commit `73016e0`): Stage-A reduction tree, with the address copies kept; earlier flow.

Depth	Mode	Cells	FFs	Area (μm^2)	FF share of area	Fmax (MHz)	WNS 1 GHz (ns)	WNS 500 MHz (ns)	Viol. 1G / 500M
16	0	10,834	2,957	26,364	0.597	626	−0.597	met	0 / 0
16	1	11,242	2,957	26,634	0.591	612	−0.635	met	0 / 0
64	0	41,192	10,431	96,531	0.575	486	−1.056	−0.308	8 / 7
64	1	40,434	10,431	96,896	0.572	439	−1.276	−0.349	9 / 5
256	0	144,237	40,225	359,697	0.595	214	−3.674	−2.674	118 / 118
256	1	146,419	40,225	361,999	0.591	199	−4.014	−3.014	48 / 48

Summary. The final design reaches 636, 603 and 557 MHz at depths 16, 64 and 256 in mode 0, and 660, 611 and 542 MHz in mode 1. The 500 MHz netlists meet 500 MHz at depths 16 and 64; at depth 256 they miss by 0.036 ns (mode 0) and 0.002 ns (mode 1), far inside the 8.53% and 7.02% noise spread at that depth, so meeting 500 MHz at depth 256 is not demonstrated either way. Flip-flops take 0.578–0.593 of the cell area in the final design (0.574–0.597 in the baseline).

Table 4: P5d (tag `rtl-v2-final`, commit `7705a7b`): hit-index priority tree; earlier flow.

Depth	Mode	Cells	FFs	Area (μm^2)	FF share of area	Fmax (MHz)	WNS 1 GHz (ns)	WNS 500 MHz (ns)	Viol. 1G / 500M
16	0	11,529	2,957	26,733	0.589	571	−0.751	met	0 / 0
16	1	11,661	2,957	26,875	0.585	587	−0.704	met	1 / 1
64	0	37,720	10,431	92,970	0.597	528	−0.893	−0.179	2 / 3
64	1	37,901	10,431	93,294	0.595	471	−1.124	−0.340	3 / 3
256	0	143,761	40,225	359,442	0.595	451	−1.216	−0.343	120 / 120
256	1	145,730	40,225	362,935	0.590	435	−1.297	−0.337	108 / 131

Table 5: P5e (same RTL, `rtl-v2-final`; flow commit `b69a931`): timing-driven ABC flow. These are the current numbers of the controller.

Depth	Mode	Cells	FFs	Area (μm^2)	FF share of area	Fmax (MHz)	WNS 1 GHz (ns)	WNS 500 MHz (ns)	Viol. 1G / 500M
16	0	11,466	2,957	26,744	0.588	636	−0.572	met	1 / 3
16	1	11,057	2,957	26,627	0.591	660	−0.516	met	1 / 1
64	0	37,939	10,431	94,386	0.588	603	−0.659	met	7 / 9
64	1	38,692	10,431	95,101	0.583	611	−0.636	met	6 / 8
256	0	149,620	40,225	369,530	0.579	557	−0.797	−0.036	61 / 100
256	1	150,254	40,225	370,227	0.578	542	−0.844	−0.002	19 / 23

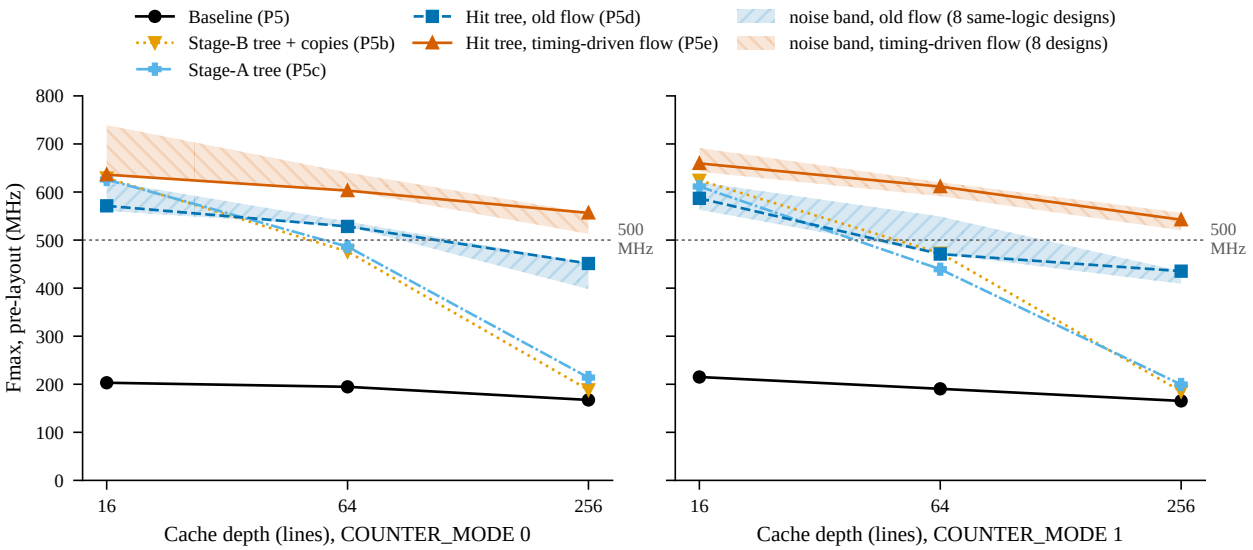


Figure 2: Pre-layout Fmax of the 1 GHz netlists vs. depth for each design iteration, with the flow noise bands (min–max Fmax of 8 designs with identical logic) for the earlier and the timing-driven flow. Script `paper/v2/figs/fig_fmax.py` (the paper's figure at report width).

6 Critical paths

Worst setup path of the 1 GHz netlists, mode 0, for the baseline and the final design at depths 16 and 256. **Tool output** (OpenSTA `report_checks`, per-cell rows): start and end registers, number of cells, arrival, slack, buffers, per-cell-type delays. The register names were mapped from flip-flop instances to RTL registers by `tools/p5_critical_path.py` using the netlists. **My reading** (not tool output): which RTL structure the path belongs to, and whether it can occur in one clock cycle.
Start → end registers (tool output):

- Baseline, depth 16: `freq_cnt[1][0]` → `valid_bits[9]`.
- Baseline, depth 256: `req_addr_reg[4]` → `freq_cnt[1][0]`.
- Final, depth 16: `freq_cnt[9][1]` → `victim_reg[0]`.
- Final, depth 256: `freq_cnt[207][1]` → `stageA_min[12][1]`.

Table 6: Critical-path summary (tool output). Cells include the launching flip-flop. Delays in ns; slack at 1 GHz.

Path	Cells	Arrival	Slack	Buffers count / delay	Longest buffer run cells / delay	Non-buffer delay
Baseline, depth 16	114	4.879	−3.923	10 / 0.486	1 / 0.036	4.393
Baseline, depth 256	87	5.934	−4.975	58 / 4.358	52 / 3.933	1.576
Final, depth 16	26	1.537	−0.572	4 / 0.145	1 / 0.046	1.392
Final, depth 256	29	1.760	−0.797	3 / 0.207	1 / 0.067	1.553

Largest delay contributions by cell type (tool output; count and summed delay in ns, six largest):

- Baseline, depth 16: AOI22_X1 32×, 1.407; OAI22_X1 26×, 0.725; MUX2_X1 10×, 0.644; OAI21_X1 8×, 0.351; AOI21_X1 4×, 0.247; BUF_X2 5×, 0.219.
- Baseline, depth 256: BUF_X1 57×, 4.268; NOR4_X1 4×, 0.472; OAI21_X1 6×, 0.189; OR4_X1 1×, 0.135; AOI21_X1 6×, 0.123; NOR3_X1 1×, 0.121.
- Final, depth 16: AOI221_X1 2×, 0.229; OAI221_X1 3×, 0.176; MUX2_X1 2×, 0.137; OAI33_X1 1×, 0.128; DFFR_X1 1×, 0.098; OAI221_X4 1×, 0.093.
- Final, depth 256: AOI221_X1 4×, 0.282; MUX2_X1 3×, 0.222; BUF_X1 3×, 0.207; AND2_X1 3×, 0.154; OR4_X1 1×, 0.118; NAND4_X1 2×, 0.112.

What the paths are (my reading of the netlist against the RTL, not tool output):

- *Baseline, depth 16*: Stage-B victim selection in EVICT, then the write that clears the victim's valid bit. In the baseline, Stage B was a serial loop over the 16 group minima; with 16 groups at depth 16 each group has one line, so the whole minimum search is this serial compare-and-select chain (the repeated AOI22/OAI22 pairs and MUX2 cells).
- *Baseline, depth 256*: the hit path in LOOKUP: registered request address → tag compare → priority loop for the hit index → counter increment. Most of its delay is a serial run of 52 buffers that ABC's buffering built on the high-fanout address net; the logic itself is short.
- *Final, depth 16*: victim selection into the victim register (Stage-B tree; at depth 16 Stage A has one line per group). The counters change only on edges into IDLE, and the victim register loads only in EVICT, at least three cycles later, so this path cannot be exercised in one cycle (Section 9).
- *Final, depth 256*: counter → Stage-A minimum register (the Stage-A tree, registered at the end of LOOKUP). The counters change only on edges into IDLE, and the Stage-A register loads at the end of

LOOKUP, at least two cycles later, so this is also not a one-cycle path. With the multicycle exceptions the worst path becomes a Stage-A register → valid bit through Stage B (tool output of the cross-check run).

7 Flow noise floor

Eight designs with identical logic were synthesized with each flow at 1 GHz: the final RTL, a flat-vector ("packed") form of the same hit tree, and three one-line rewrites of each (swapped operands of the tag compare and of the hit-tree OR, and a comparison written the other way round; `tools/p5d_null_variants.py`). Any spread in their Fmax is variation of the flow, not of the design. Spread = max/min − 1.

		Old flow (P5d)		Timing-driven flow (P5e)	
Depth	Mode	Fmax min–max (MHz)	Spread	Fmax min–max (MHz)	Spread
16	0	561–623	11.12%	636–739	16.23%
16	1	563–621	10.20%	642–692	7.71%
64	0	528–539	1.96%	603–641	6.24%
64	1	471–549	16.62%	591–621	4.92%
256	0	398–452	13.54%	513–557	8.53%
256	1	409–435	6.42%	520–557	7.02%

8 Verification

RTL. A self-checking testbench with 7 tests passes at every depth and mode. Each RTL version was compared in lockstep with the earlier versions on random traffic: every run in the eight lockstep result files of P5b–P5d (six runs each, 200,000 cycles) passed with 0 port and 0 victim-slot mismatches, and 0 hit-index mismatches in the P5d files, the only ones that record them. Mutation testing of the final RTL: 113 mutant runs killed by the unit tests, 10 more by the lockstep check, 33 runs of equivalent mutants; result "OK (all non-equivalent mutants killed)".

Cell models. The gate-level simulation uses cell models generated by Yosys from the liberty file. They were checked against the liberty functions by exhaustive truth tables (combinational cells) and random stimulus (flip-flops): 84 cells pass, 1 fails. The failing cell is DFFRS_X1 (flip-flop with asynchronous clear and preset), whose generated model differs from the liberty function when clear and preset are both active and when one of them is released while the other is still active; no netlist uses DFFRS_X1. The netlists use 84 distinct cell types, all with a model that passed; there are no scan flip-flops.

Gate-level lockstep (functional, zero delay; Icarus 11.0 on the cluster). Each of the 12 netlists runs the same 200,000 random cycles as the RTL; hit flag, response valid, response data, fill request and fill address are compared every cycle. Result: 0 mismatching cycles and no X on any output in all 12. An RTL rerun of the same stimulus shows that it drives counters to their maximum of 63 in every depth and mode (`results/P7/counter_sat/`).

Depth	Mode	MHz	Cycles	Hits	Misses	Mismatch	X (gl/rtl)	vvp wall	Peak RSS (kB)
16	0	1000	200000	57297	8139	0	0/0	7:53.04	65392
16	0	500	200000	57297	8139	0	0/0	8:16.61	65472
16	1	1000	200002	57090	8166	0	0/0	8:11.14	64680
16	1	500	200002	57090	8166	0	0/0	7:43.67	64860
64	0	1000	200000	57092	8199	0	0/0	25:07.54	212308
64	0	500	200000	57092	8199	0	0/0	25:11.24	212688
64	1	1000	200000	53588	9369	0	0/0	25:21.85	215572
64	1	500	200000	53588	9369	0	0/0	24:32.00	215580
256	0	1000	200000	57209	8109	0	0/0	1:37:59.46	819684
256	0	500	200000	57209	8109	0	0/0	1:39:11.79	819572
256	1	1000	200000	56927	8216	0	0/0	1:40:56.35	825804
256	1	500	200000	56927	8216	0	0/0	1:40:07.57	825892

Unit tests on the netlists. The 7-test testbench, unmodified, on the depth-16 and depth-64 netlists with Icarus 12.0 (local) and on the depth-16 and depth-256 netlists with Icarus 11.0 (cluster): 16 runs, all pass.

Depth	Mode	MHz	Where	Result	Compile	Run
16	0	1000	local	pass	5.14 s	32.94 s
16	0	500	local	pass	3.32 s	30.20 s
16	1	1000	local	pass	3.58 s	28.74 s
16	1	500	local	pass	3.63 s	29.21 s
64	0	1000	local	pass	73.16 s	96.20 s
64	0	500	local	pass	57.07 s	98.71 s
64	1	1000	local	pass	54.71 s	112.18 s
64	1	500	local	pass	56.23 s	113.59 s
16	0	1000	cluster	pass	0:01.92	0:11.67
16	0	500	cluster	pass	0:02.03	0:11.81
16	1	1000	cluster	pass	0:01.93	0:11.98
16	1	500	cluster	pass	0:02.01	0:11.93
256	0	1000	cluster	pass	13:03.75	7:02.92
256	0	500	cluster	pass	13:09.11	6:57.05
256	1	1000	cluster	pass	9:45.77	6:41.02
256	1	500	cluster	pass	9:40.19	6:39.94

Equivalence (Yosys `equiv_make`, `equiv_simple`, `equiv_induct`). The four depth-16 netlists are proven equivalent to the RTL: 2977 of 2977 equivalence points, wall time 463 to 698 s each. Caveats: the asynchronous reset is treated as synchronous (`async2sync`); the induction starts from matched registers being equal; depths 64 and 256 were not attempted with the netlists.

9 Known issues and open questions

- **ABC script not ablated.** The timing-driven script was chosen on one pilot (depth 16, mode 0); which of its parts helps is unknown.
- **Area and violations.** Against the earlier flow on the same RTL, it uses 1.52% and 1.94% more area at depth 64 and 2.81% and 2.01% more at depth 256 (modes 0 and 1), and slew/capacitance violators at depth 64 rise from 2 to 7 and 3 to 6 (they fall at depth 256, from 120 to 61 and 108 to 19). A layout flow would buffer these nets.
- **Flip-flop storage.** Tags, vectors, counters and valid bits are flip-flops (0.578–0.593 of the cell area in the final design); an SRAM macro would change area and timing and was not available in this flow.

- **Depth 256 at 500 MHz** misses by less than the noise (Section 5); not demonstrated.
- **Paths that may not be exercisable.** Reading the FSM, three register pairs cannot form one-cycle paths: Stage-A register → data and tag arrays (2 cycles), counters → Stage-A register (2), counters → valid bits and victim register (3) (`tools/p5e_mcp.py`). As OpenSTA multicycle exceptions they change WNS by at most 0.063 ns (earlier flow) and 0.026 ns (current flow); at depth 256 and 500 MHz the current flow stays at −0.0357 ns (mode 0) and reaches −0.0001 ns (mode 1). The headline numbers use no exceptions.

			Old flow WNS (ns)		Timing-driven WNS (ns)	
Depth	Mode	MHz	without	with	without	with
16	0	1000	−0.751	−0.719	−0.572	−0.554
16	0	500	0	0	0	0
16	1	1000	−0.704	−0.694	−0.516	−0.490
16	1	500	0	0	0	0
64	0	1000	−0.893	−0.893	−0.659	−0.652
64	0	500	−0.179	−0.179	0	0
64	1	1000	−1.124	−1.096	−0.636	−0.635
64	1	500	−0.340	−0.277	0	0
256	0	1000	−1.216	−1.216	−0.797	−0.792
256	0	500	−0.343	−0.343	−0.036	−0.036
256	1	1000	−1.297	−1.291	−0.844	−0.844
256	1	500	−0.337	−0.326	−0.002	−0.0001

- **RTL vs. the trace simulator.** The trace simulator that motivates the controller breaks LFU ties by the oldest entry and caps counts at 255; the RTL breaks ties by the lowest slot index and saturates at 63. The RTL was not replayed on the Criteo traces.

10 What a real signoff would need

Future work, stated without any claim about its outcome: a licensed foundry PDK and standard-cell library (for example TSMC, under NDA) instead of NanGate45; SRAM macros for the tag and data arrays; placement, clock-tree synthesis and routing; parasitic extraction and timing with extracted parasitics across process corners; a reset distribution tree; power analysis with realistic switching activity; gate-level simulation with SDF back-annotation; and equivalence checking at every depth.

11 Appendix: file index and regeneration commands

Result files (commit IDs of the current history; old IDs before the history rewrite are translated in `results/COMMIT_MAP.md`).

File	Content	Added in	Last changed
results/P5/RESULTS.md	P5 baseline, all runs	b5d85b7	b5d85b7
results/P5/SOURCES.md	toolchain and liberty record	b5d85b7	b5d85b7
results/P5/CRITICAL_PATH.md	baseline critical paths	99cdde6	0d66a3a
results/P5b/RESULTS.md	P5b runs	4debe84	73016e0
results/P5c/tree_copies/RESULTS.md	P5c runs (kept variant)	73016e0	73016e0
results/P5c/tree_only/RESULTS.md	P5c tree only	73016e0	73016e0
results/P5d/RESULTS.md	P5d runs (rtl-v2-final, old flow)	7705a7b	7705a7b
results/P5d/NOISE.md	old-flow noise floor	7705a7b	7705a7b
results/P5d/mutation.txt	mutation test of the final RTL	7705a7b	7705a7b
results/P5e/new/RESULTS.md	P5e runs (final)	b69a931	b69a931
results/P5e/COMPARE.md	flow comparison, noise, multicycle	b69a931	b69a931
results/P5e/CRITICAL_PATH.md	final critical paths	b69a931	b69a931
results/P5f_gls/CELLCHECK.md	cell-model check	4188580	4188580
results/P5f_gls/COVERAGE.md	cell-type coverage	4188580	4188580
results/P5f_gls/equiv/	depth-16 equivalence	42a8201	42a8201
results/P5f_gls/cluster/	cluster gate-level runs	02518be	02518be
results/P5f_gls/sim/	local gate-level unit tests	4188580	4188580
results/P5f_gls/SUMMARY.md	gate-level summary	02518be	061f0cf
results/P7/TABLES.md	all tables, generated	1575c09	f0f82ce
results/COMMIT_MAP.md	old-to-new commit IDs	061f0cf	061f0cf

Commands (from the repository root; the liberty file and OpenSTA as in results/P5/SOURCES.md; local machine unless noted). The run scripts committed at the historical commits name their reference commits by IDs from before the history rewrite; translate them with results/COMMIT_MAP.md (the current versions of the scripts already use the new IDs).

```
# P5 baseline (tag rtl-v2-baseline) and P5b, earlier flow
git worktree add ../base rtl-v2-baseline
(cd ../base && bash synthesis/p5/run_p5.sh)
git worktree add ../p5b 4debe84
(cd ../p5b && OUTDIR=results/P5b bash synthesis/p5/run_p5.sh)
# P5c (both variants), P5d (rtl-v2-final, incl. noise floor), earlier flow
git worktree add ../p5c 73016e0
(cd ../p5c && bash synthesis/p5c/run_p5c.sh)
bash synthesis/p5d/run_p5d.sh
# P5e: timing-driven flow, noise floor, multicycle cross-check, report tables
bash synthesis/p5e/run_p5e.sh
# RESULTS.md of one phase (needs the liberty for the flip-flop areas)
python3 -I tools/p5_results.py results/P5e/new \
    ~/eda/pdk/nangate45/NangateOpenCellLibrary_typical.lib
# critical-path tables
python3 -I tools/p5_critical_path.py results/P5e/new d16_m0_1000MHz d256_m0_1000MHz
# P5f: netlists, cell models, checks, local tests, equivalence; cluster jobs
bash synthesis/p5f/regen_netlists.sh
bash synthesis/p5f/run_p5f.sh models cellcheck coverage sim equiv
bash synthesis/p5f/submit_gls.sh          # on the cluster login node
# all tables, this report's tables, the number check, the figure and the PDF
python3 -I tools/make_tables.py
python3 -I docs/synthesis_report/gen_tables.py
python3 -I tools/paper_numbers_check.py docs/synthesis_report
.venv/bin/python paper/v2/figs/fig_fmax.py docs/synthesis_report/figs/fig_fmax.pdf 6.5
cd docs/synthesis_report && latexmk -pdf synthesis_report.tex
```